Multimodal Models for Segmentation via Textual Prompts

Viktoriia Zinkovich

In the first lecture on Multimodal Large Language Models (MLLMs), we primarily considered Vision–Language Models (VLMs), which take image and text as input and produce textual output, i.e., image + text → text. However, models for **Referring Expression Segmentation (RES)** belong to a different class: they generate a segmentation mask conditioned on both the image and a textual description, i.e., image + text → image (segmentation mask), as illustrated in Figure 1.



Figure 1: **Examples of reasoning-based segmentation.** Given an image and a textual query, the model predicts a pixel-wise mask corresponding to the described object.

As shown in Figure 1, such models go beyond pure language generation and require grounding textual semantics into spatial predictions. In this part of the homework, you will explore these models and develop an intuition for their behavior.

Contents

1 Referring Expression Segmentation **2**
1.1 Practical Task . . . 2
1.2 Conclusions (spoilers!) . . . 2

2 Reasoning Segmentation **3**
2.1 Practical Task . . . 3
2.2 Conclusion . . . 4

1 Referring Expression Segmentation

A natural starting point for exploring this field is benchmarking platforms such as `PapersWithCode`. For RES, a widely used benchmark is **RefCOCO**. Models are evaluated using the Intersection-over-Union (IoU) metric on validation and test splits, as shown in Figure 2.

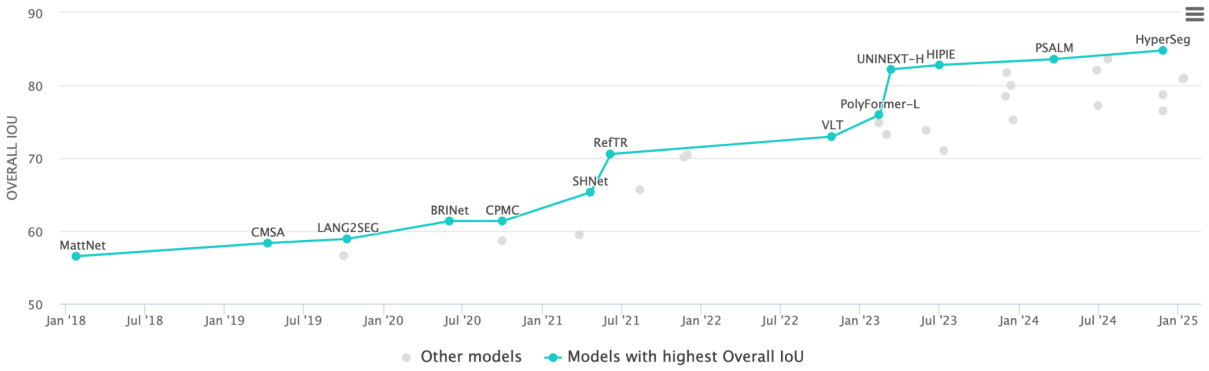


Figure 2: **Evolution of models on the RefCOCO benchmark.** Performance is measured using IoU, with recent models achieving steadily improving results.

As illustrated in Figure 2, modern approaches achieve strong performance on this benchmark. To ground our exploration, we consider the **HIPIE model** [1] (see Figure 3), a high-performing method published at NeurIPS 2023. HIPIE offers a practical trade-off between performance and usability: while slightly below the state-of-the-art (oIoU 82.8 vs. 84.8 for HyperSeg [2]), it remains significantly easier to deploy in our setting.

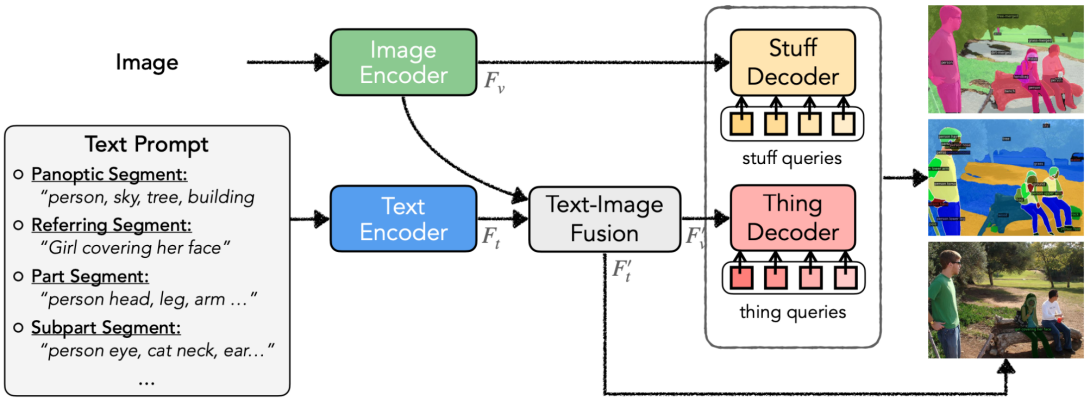


Figure 3: **HIPIE architecture** [1]. A unified framework for referring expression segmentation that integrates visual and textual features for mask prediction.

1.1 Practical Task

Proceed to the notebook and complete the implementation of the provided functions. It is recommended to run the notebook on Kaggle using a T4 GPU to avoid memory issues or environment inconsistencies. Once you have finished, you may return here to gain further insights.

1.2 Conclusions (spoilers!)

It is recommended to read this section only after completing the practical task.

A key observation from the above experiments is that many state-of-the-art RES models trained on RefCOCO tend to segment *non-existent objects*. This behavior arises because the dataset primarily contains short and simple referring expressions, rarely including negative or ambiguous cases.

To address this limitation, the **SESAME model** [3] introduces the FP-RefCOCO(+/g) benchmark, which incorporates *negative prompts* describing objects that are absent from the image. As illustrated in Figure 4, this enables models to learn when *not* to segment.

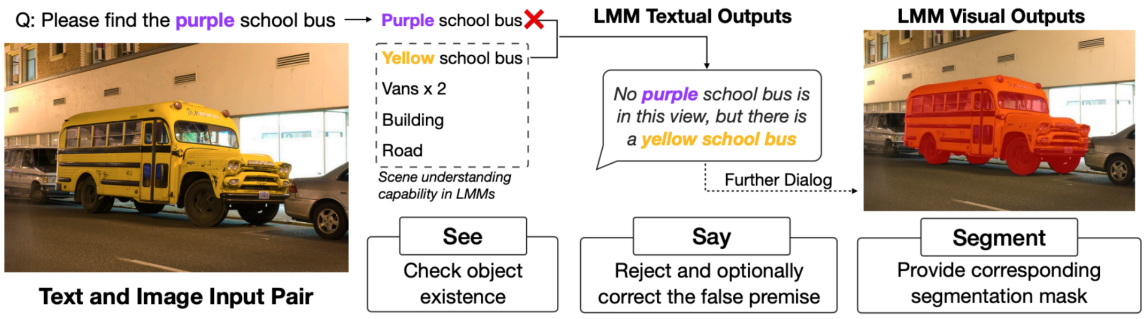


Figure 4: **SESAME architecture** [3]. The model incorporates mechanisms to handle negative prompts and avoid false-positive segmentations.

Since FP-RefCOCO differs from the standard benchmark, it is not always listed on common leaderboards such as `PapersWithCode`, which may initially cause confusion.

2 Reasoning Segmentation

Beyond RefCOCO-style benchmarks, a new line of work has emerged focusing on **reasoning-based segmentation**. Instead of simple referring expressions, these models handle queries requiring world knowledge and reasoning.

The **LISA model** [4] introduces the *Reasoning Segmentation* task and the ReasonSeg benchmark. In this setting, queries may take forms such as *"fruit with the highest vitamin C"* rather than explicit object names. Figure 5 illustrates the architecture of LISA, which serves as a foundation for subsequent works such as VISA [5], GSVA [6], and CoReS [7].

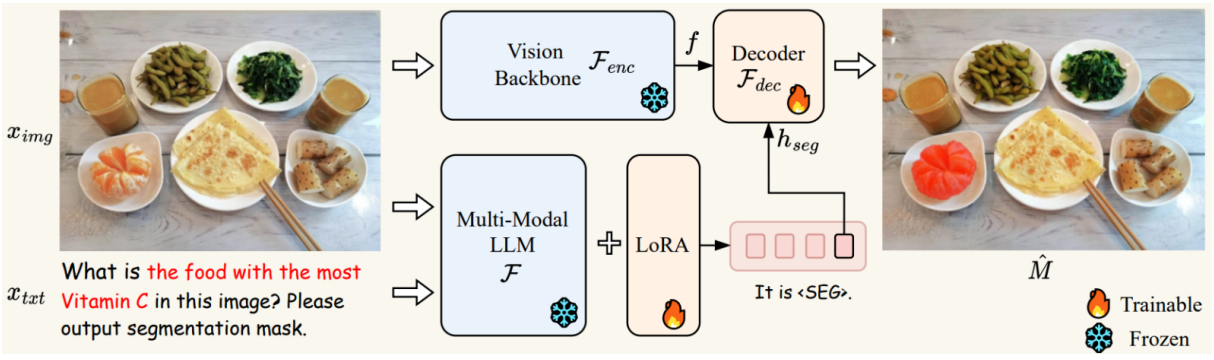


Figure 5: **LISA architecture** [4]. The model combines LLM reasoning with segmentation via token-based mask generation.

2.1 Practical Task

Return to the notebook and complete the remaining implementations.

Important:

- Restart the notebook from scratch (do not forget to **save outputs of the cells above**),
- Save intermediate outputs,
- Ensure reproducibility of results.

Again, using Kaggle with a T4 GPU is strongly recommended.

2.2 Conclusion

Congratulations on completing this homework!

You dived deeply into vision-language models for object segmentation given textual prompts. At the beginning, you examined the classic RefCOCO benchmark with short, simple prompts and state-of-the-art models for it. Due to the simplicity of the benchmark, you saw that such models (HIPIE, GLEE-Pro, UniLSeg, UNINEXT...) tend to segment non-existent objects and are not capable of reasoning (Section 1).

Further, we looked at models aimed at solving this problem by explicitly refusing to segment objects that are not present in the image. For example, this can be achieved through an additional LLM that checks whether such an object exists in the image before initiating the segmentation process (SESAME) or by introducing a special `<REJ>` token (GSVA [6]).

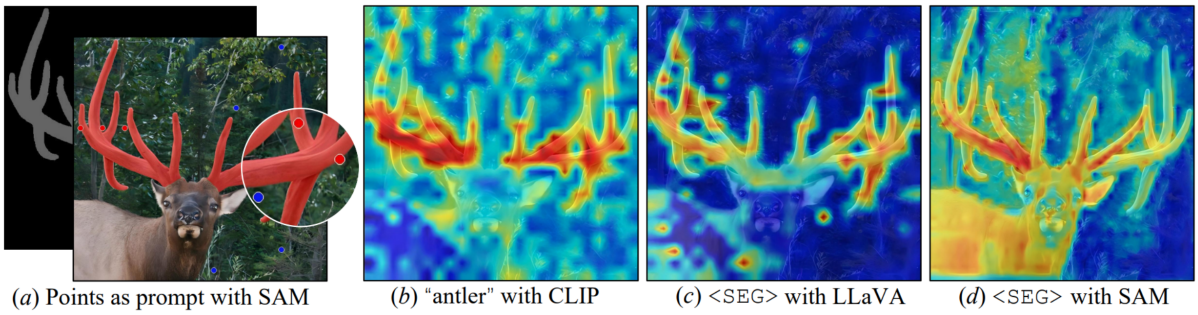


Figure 6: **Token-based segmentation decoding.** The `<SEG>` token embedding is transformed into a segmentation mask, which can be further refined by downstream modules.

Finally, we examined the foundation model LISA (Figure 5), which first proposed the task of Reasoning segmentation with more intricate prompts and complex world knowledge. In the proposed LISA model, that employs an embedding-as-mask approach where a `<SEG>` token in the LLM vocabulary is decoded via SAM to generate segmentation mask. As you could see, the purpose of the token is to solve the sent puzzle during its time in the LLM (for example, "Who was the US president in 2010?"). Once the puzzle is solved, the resulting embedding of this token, which actually reflects the mask of the solved object, is fed into SAM only to decode and refine the required mask.

Thus, in the paper "Reasoning to Attend: Try to Understand How Token Works" [8], it is shown that at this stage, it's even possible to improve SAM's predictions, knowing that the mask is already prepared and only needs to be refined.

References

[1] Xudong Wang, Shufan Li, Konstantinos Kallidromitis, Yusuke Kato, Kazuki Kozuka, and Trevor Darrell. Hierarchical open-vocabulary universal image segmentation, 2023.

[2] Cong Wei, Yujie Zhong, Haoxian Tan, Yong Liu, Zheng Zhao, Jie Hu, and Yujiu Yang. Hyperseg: Towards universal visual segmentation with large language model, 2024.

[3] Tsung-Han Wu, Giscard Biamby, David Chan, Lisa Dunlap, Ritwik Gupta, Xudong Wang, Joseph E. Gonzalez, and Trevor Darrell. See, say, and segment: Teaching lmms to overcome false premises, 2023.

[4] Xin Lai, Zhuotao Tian, Yukang Chen, Yanwei Li, Yuhui Yuan, Shu Liu, and Jiaya Jia. Lisa: Reasoning segmentation via large language model, 2024.

[5] Cilin Yan, Haochen Wang, Shilin Yan, Xiaolong Jiang, Yao Hu, Guoliang Kang, Weidi Xie, and Efstratios Gavves. Visa: Reasoning video object segmentation via large language models, 2024.

[6] Zhuofan Xia, Dongchen Han, Yizeng Han, Xuran Pan, Shiji Song, and Gao Huang. Gsva: Generalized segmentation via multimodal large language models, 2024.

[7] Xiaoyi Bao, Siyang Sun, Shuailei Ma, Kecheng Zheng, Yuxin Guo, Guosheng Zhao, Yun Zheng, and Xingang Wang. Cores: Orchestrating the dance of reasoning and segmentation, 2024.

[8] Rui Qian, Xin Yin, and Dejing Dou. Reasoning to attend: Try to understand how <seg> token works, 2026.